

Multiple Files Executed Under One Spool

Oracle SQL*Plus & SQLcl Script Management Guide

To run multiple separate SQL script files under a single master spool file, you use a **Master Script** that starts spooling, uses the @ or START command to run each sub-script sequentially, and then turns spooling off.

1. Create Your Sub-Scripts

01_create_user.sql

```
-- Executed as SYSDBA
CREATE USER dev_user IDENTIFIED BY "Password123#"
  DEFAULT TABLESPACE users
  QUOTA UNLIMITED ON users;

GRANT CREATE SESSION, CREATE TABLE TO dev_user;
```

02_create_tables.sql

```
-- Executed as dev_user
CREATE TABLE departments (
  dept_id NUMBER PRIMARY KEY,
  dept_name VARCHAR2(50) NOT NULL
);

CREATE TABLE employees (
  emp_id NUMBER PRIMARY KEY,
  emp_name VARCHAR2(50),
  dept_id NUMBER REFERENCES departments(dept_id)
);
```

03_insert_data.sql

```
-- Executed as dev_user
INSERT INTO departments VALUES (10, 'IT');
INSERT INTO departments VALUES (20, 'HR');

INSERT INTO employees VALUES (101, 'Alice', 10);
INSERT INTO employees VALUES (102, 'Bob', 20);

COMMIT;
```

2. Create the Master Driver Script (run_all.sql)

This script acts as the controller. It opens the single spool log and calls each individual script in sequence using @.

```

-- Setup display settings and start logging everything into ONE file
SET ECHO ON;
SET SERVEROUTPUT ON;
SPOOL master_execution_log.txt REPLACE;

-- Step 1: Create user and permissions (as SYSDBA)
PROMPT *** EXECUTING STEP 1: USER CREATION ***
@01_create_user.sql

-- Step 2: Switch to the newly created user session
PROMPT *** CONNECTING AS NEW USER ***
CONNECT dev_user/"Password123#";

-- Step 3: Run table creation script
PROMPT *** EXECUTING STEP 2: TABLE CREATION ***
@02_create_tables.sql

-- Step 4: Run data insertion script
PROMPT *** EXECUTING STEP 3: DATA INSERTION ***
@03_insert_data.sql

-- Step 5: Verification Query
SELECT e.emp_id, e.emp_name, d.dept_name
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id;

-- Stop logging to the single spool file
SPOOL OFF;

```

3. Execution

Ensure run_all.sql, 01_create_user.sql, 02_create_tables.sql, and 03_insert_data.sql are stored in the same folder.

- Open SQL*Plus/SQLcl and log in as SYSDBA:

```
sqlplus sys/your_sys_password as sysdba
```

- Execute the master script:

```
SQL> @run_all.sql
```